

Practice Questions

Test your comprehension by answering the following questions:

1. What are nodes and edges?
2. What are root and leaf nodes?
3. What are the three tree traversal orders?
4. What does *DAG* stand for?
5. What is a cycle, and do DAGs have cycles?
6. What is a binary tree?
7. What are the child nodes in a binary tree called?
8. If a parent node has an edge to a child node, and the child node has an edge back to the parent node, is this graph considered a DAG?
9. What is backtracking in a tree traversal algorithm?

For the following tree traversal problems, you can use the Python/JavaScript code in “A Tree Data Structure in Python and JavaScript” in Chapter 4 for your tree and the multiline `MAZE` string from the *mazeSolver.py* and *mazeSolver.html* programs for the maze data.

10. Answer the three questions about recursive solutions for each of the recursive algorithms presented in this chapter:
 - a. What is the base case?
 - b. What argument is passed to the recursive function call?
 - c. How does this argument become closer to the base case?

Then re-create the recursive algorithms from this chapter without looking at the original code.